



The impact of dynamic learning and training on the personnel staffing decision

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ABSTRACT

This paper studies the personnel staffing decision accounting for the planning of workforce training and the inclusion of learning and forgetting, which dynamically change the workforce skill level, influencing the required staffing budget. This contrasts staffing models presented in literature, which determine the required number of workers under the assumption of a fixed and/or predefined skill mix of the workforce. We consider different types of on-the-job and off-the-job training activities, traditional learning associated with execution of regular tasks and forgetting according to which a worker's efficiency is adapted dynamically throughout the time horizon. The acquired worker's efficiency level determines the speed at which assigned tasks can be processed, which is modelled via the definition of multiple discrete modes per task. To improve the accuracy of the staffing decision and account for the effect of gradually changing the workers' skill levels, a baseline schedule is composed that comprises the solving of an integrated days-off scheduling and task assignment problem. We propose a dedicated solution procedure that relies on a constructive heuristic to find an initial solution and subsequently a branch-and-price algorithm to derive a (near-)optimal solution. The branch-and-price algorithm is featured by a multi-stage pricing procedure to find promising columns, alternating between mathematical mixed-integer programming, a single-pass heuristic and a genetic algorithm. The computational experiments validate the performance of the proposed procedure and demonstrate the benefits of different training and learning methods under different conditions.

1. Introduction

Adequate management of personnel is of crucial importance for many companies. As labour costs comprise one of the biggest cost components for firms, even a marginal decrease in cost resulting from a more efficient use of personnel via proper planning or scheduling could result in huge benefits (Van den Bergh et al., 2013). Moreover, due to the shortage of (qualified) personnel, it is of crucial importance to satisfy the task demand as efficiently as possible with no more personnel than minimum required and/or provide training whenever required. In this context, the personnel staffing problem is highly relevant, which is the focus of the presented study. The vast majority of research, however, has focused either on the staffing decision given a predefined skill mix or on the budgeting and planning of required training, but neglects the consequences of learning and training of workers on the staffing decision and the associated dynamic changes to the workers' skill levels (De Bruecker et al., 2015).

In this paper, we study the impact of different skill improvement methods on the personnel staffing decision. The staffing budget is

accurately determined based on the construction of a baseline personnel days-off and task schedule to account for complicating scheduling constraints and the effect of a gradually changing skill mix of the workforce throughout the planning horizon. The underlying baseline scheduling problem is a combined days-off scheduling and task assignment problem. The days-off scheduling problem determines the line-of-work of a worker by assigning a day off or a working day on each day of the planning horizon (Morris & Showalter, 1983). The task assignment problem involves the scheduling of training activities and allocation of regular tasks to workers based on their availability, which is determined by their days-off lines-of-work. These two sequential problems of days-off scheduling and personnel task assignment are integrated for following reasons. The skills of each worker are expressed by a competence score that is linked with a specific efficiency value via a transformation function. The non-linear learning curves are approximated via a discrete function, defining discrete task operation modes. The efficiency level determines the duration to complete a task based

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on the selection of an appropriate task mode, and is determined by both the individual task and days-off schedule of a worker and his/her prevailing skill level at the time of executing the task. In this way, we can consider learning and/or (scheduling of) training activities, potentially improving the required personnel budget.

There are different ways to adjust the skill mix of the workforce during the execution of a schedule. The first way is *learning and forgetting*. The learning effect is a phenomenon that occurs when personnel gains larger experience and a higher skill level by doing previously assigned tasks, reducing the time required to complete other related upcoming tasks (Mosheiov, 2001). However, personnel may suffer from *forgetting*, leading to a decrease in skill level and productivity after some time period (e.g. Nembhard & Osothsilp, 2001). Secondly, we distinguish between two broad categories of training to increase a worker's efficiency in a more formal manner, i.e. (i) on-the-job or workplace training, for which there is a direct link with the company and workers acquire company-specific knowledge relative to tools, equipment, machines and documents; and (ii) off-the-job or external training, for which the workers are away from the company such that they can gather occupation-specific knowledge (Gaulke, 2021; Vasanthi & Basariya, 2019). Off-the-job training is typically more costly and takes a somewhat longer duration, but leads to new insights and, consequently, a larger increase in the skill level of a worker, inducing a larger decrease in the time required to execute related tasks. The most prominent difference between training and learning is that the former should be planned explicitly by training tasks and possibly involves a tutor, while the latter is realised single-handedly by completing regular tasks. When each worker is assigned to an appropriate set of (training) tasks, a single worker may be able to process a larger number of tasks, which could potentially reduce the number of required resources to cover all tasks. These effects of training, learning and forgetting have been neglected in the staffing models from literature. However, the potential benefits in staffing budget resulting from any learning method depend upon environmental conditions, the learning policy and the extent to which tasks have similar characteristics. The main research objectives of this study are to (i) study the impact of training, learning and forgetting on the staffing decision and propose a proper methodology, and (ii) investigate under which conditions, which type of learning is more relevant to consider from a staffing objective.

The personnel staffing problem under study is complicated due to the dynamically changing skill level of workers and the definition of multiple modes per task having a specific duration accounting for the workers' skill level to process the task. Accordingly, we propose a dedicated procedure to retrieve high-quality solutions that takes instance-specific characteristics into account to improve the computational efficiency. The procedure is initialised by a constructive heuristic that composes for each individual worker a line-of-work covering as many tasks as possible. To reduce the complexity of the problem and the presence of symmetries, some of the lines-of-work with the largest number of tasks included and corresponding task-worker assignments may be fixed. The remaining tasks and required workers are scheduled by applying a branch-and-price procedure, which relies on column generation (CG) and branching to find a high-quality integer solution. To accelerate the CG procedure, we propose a multi-stage pricing that encompasses three stages according to which multiple promising lines-of-work may be constructed. When branching is required to yield an integer solution, a dedicated follow-on task branching strategy is implemented to break symmetries. A lower bound is calculated accounting for the highest possible efficiency level of workers to enable the early termination of the algorithm. To validate the procedure, comparison is made with solving the original mixed-integer programming (MIP) formulation via a commercial solver and the proposed constructive heuristic.

The remainder of this paper is organised as follows. Section 2 gives an overview of previous studies related to personnel staffing and scheduling. In addition, we discuss the existing literature concerning

the incorporation of training into scheduling problems, and definition of learning and forgetting. The technical details relative to adapting workers' skill levels resulting from training, learning and forgetting can be found in Section 3. Section 4 describes the personnel staffing problem, integrating days-off scheduling and task assignment in view of different methods to gradually adapt the skill mix of the hired workforce. The proposed solution procedure is presented in Section 5. In Section 6, the performance of the proposed method and impact of learning and different training methods under varying conditions are evaluated. Conclusions are given in Section 7.

2. Literature review

The personnel staffing problem has been widely investigated in the literature and can be classified in three different categories, i.e. shift scheduling, tour scheduling and days-off scheduling (Ernst et al., 2004). In this review, we focus on the latter, which assigns all workers to a line-of-work consisting of working days and days off during the planning horizon. To solve this problem, the number of required workers can be determined in an analytical manner, taking into account a limited number of time-related scheduling constraints (e.g. Alfares, 1998; Burns & Carter, 1985), or via application of mixed-integer programming (Beaumont, 1997; Caprara et al., 2003). However, when the problem size increases, it becomes harder to solve these latter optimisation models and tighter reformulations of staffing models have been proposed relying on decision variables that assign workers to complete lines-of-work (Eitzen et al., 2004). As composing all lines-of-work is intractable for larger instances, CG and related branch-and-price methods have been introduced (e.g. Dohn & Mason, 2013). In this perspective, Maenhout and Vanhoucke (2013) combine the staffing problem with the personnel scheduling problem such that a baseline schedule is composed to obtain a more accurate staffing decision.

For many application areas, the underlying personnel scheduling problem embodies the scheduling of (shift) duties, comprising a set of tasks that are paired, over the planning horizon. Krishnamoorthy and Ernst (2001) provide an overview of the personnel task scheduling problem (PTSP) and its variants. Most of these problems have been tackled using heuristic optimisation procedures due to complexity. For the single-day shift minimisation PTSP, which aims to minimise the number of workers, Smet et al. (2014) and Lin and Ying (2014) propose constructive heuristics to obtain an initial solution, which is further improved via mathematical programming. Hojati (2018) focus especially on larger instances and propose a greedy heuristic that solves a reduced problem for each individual worker in a separate manner. The study of Chandrasekharan et al. (2020) presents a constructive matheuristic that relies on the definition of smaller subproblems, which are solved in a sequential manner, by applying an employee-based and time-based decomposition.

Training can be used to improve the skills of a worker and increase his/her associated performance to complete tasks or to learn how to execute different types of tasks, which is better known as cross-training. Distinction is made between two broad categories of training, i.e. on-the-job and off-the-job training, which can be extended with self-development (Lin et al., 1996). Training of workers is typically a decision that is made in advance instead of allowing the model to autonomously decide on the most appropriate training for workers (De Bruecker et al., 2015). When on-the-job training is integrated in the scheduling problem, the decisions relative to the moment and the amount of training are considered, but they neglect the consequences of the (improved) workers' skills on the task execution. For example, Cappanera and Gallo (2004) study the airline crew rostering problem and focus not only on determining the work assignments and crew breaks, but also consider the scheduling of mandatory training periods. However, they do not account for any realised benefits resulting from these training sessions and associated skill improvements. Beddoe et al.

(2009) study the nurse rostering problem with consideration of on-the-job training, but neglect the consequences of an improved skill level. [Juang et al. \(2007\)](#) developed a training course scheduling model that concentrates only on the composition of a timetable for workers involving the training tasks. [De Bruecker et al. \(2018\)](#) are the only that account for the effect of training to compose the workforce schedule for an aircraft maintenance company. More precisely, in their study, workers need to acquire specific licenses before they can be allocated to specific types of aircraft. They propose a three-stage mixed-integer programming approach that first constitutes a shift schedule, followed by determining the required skill mix and the types of skills that need to be trained. Afterwards, they compose a schedule to decide on the timing of the training activities. To the best of our knowledge, only a few studies consider the learning from more efficient co-workers, also known as shadow training, and scheduling of involved workers. [Hosseinian et al. \(2019\)](#) are the first to study a project scheduling problem where workers are able to improve their efficiencies by learning from more efficient staff members that are assigned to the same activity. This paper models the instructing relations between workers in the form of social networks, such that for each skill and activity a network is created. Learning from a co-worker is initiated when the difference in skill level is larger than a random threshold value. [Chouhan and Goyal \(2010\)](#) model a skill capacity planning problem minimising costs in perspective of a stochastic customer demand considering hiring and skill transformation strategies. The latter involve traditional learning and induced skill transformation or shadow training where workers learn from an expert co-worker. [Ruf et al. \(2022\)](#) study a dynamic workforce capacity planning problem considering downgrading, resignation, and off-the-job and on-the-job training. The latter possibly requires an instructor for which a productivity loss is accounted due to the time spent to supervision.

The phenomenon of learning has been studied in different guises for different application domains ([Glock et al., 2019](#); [Wright, 1936](#)). There are four factors that might impact the workers' learning process, i.e. (i) the structure of training programs as autonomous learning advocates a faster learning rate ([Vits & Gelders, 2002](#)); (ii) the motivation of the worker ([Kanfer, 1990](#)); (iii) prior experience ([Nembhard & Uzumeri, 2000a, 2000b](#)); and (iv) task complexity ([Nembhard & Osothsilp, 2002](#)). In contrast to learning, forgetting received less attention in literature. Forgetting is the phenomenon where workers forget (partly) what they have learned ([Arzi & Shtub, 1997](#)). This lack of skill retention can have multiple reasons (e.g. breaks, sickness) and leads to a loss in productivity or efficiency of the workforce. Forgetting is impacted by four different factors, i.e. (i) break length ([Nembhard & Osothsilp, 2001](#)); (ii) amount of learning prior to taking a break ([Shtub et al., 1993](#)); (iii) nature of the task ([Arzi & Shtub, 1997](#); [Dar-El, 2013](#)); and (iv) types of activities performed during the interruption ([Dar-El, 2013](#)). The way in which learning and forgetting impact the worker's learning process has been modelled via curves. An overview of the most well-known and used univariate curves can be found in [Table 1](#). We distinguish three groups of models i.e. (i) log-linear models; (ii) hyperbolic models; and (iii) exponential models. The log-linear models are very popular and widely used in literature thanks to their flexibility. A comprehensive comparison between curves can be found in [Anzanello and Fogliatto \(2011\)](#) and [Jaber \(2016\)](#). Note that all these curve functions have implemented learning but only a minority considered both learning and forgetting effects.

For an exhaustive overview of the literature implementing scheduling models with consideration of traditional learning and/or forgetting, we refer to the studies of [Biskup \(2008\)](#), [Azzouz et al. \(2018\)](#) and [Paredes-Astudillo et al. \(2022\)](#). When workers perform a task repeatedly, they are able to process it more efficiently depending on the workers' experience. [Biskup \(2008\)](#) classifies existing models into two main approaches. The workers' experience and associated processing times may either depend on the number of previously processed jobs, i.e. position-based learning, or on the sum of the

processing times of jobs already processed, i.e. time-based learning. This classification has been extended with truncated and exponential learning-based functions in the review study of [Azzouz et al. \(2018\)](#). Truncated functions are proposed by [Wu et al. \(2013\)](#), amongst others, to model limits on the learning of human activity, whereas exponential functions acknowledge processing times decrease progressively and the learning activity is different depending upon the workers' learning so far. The concepts of learning and forgetting for scheduling problems have been primarily investigated in a production or project environment. The majority of studies incorporated learning effect as an influencing factor to minimise the makespan, flow time, total completion time, tardy jobs, or costs (e.g. [Van Peteghem & Vanhoucke, 2015](#); [Wu & Sun, 2006](#)). Only a few studies consider staff-related objectives, investigating the production or project effectiveness together with the objective of skill development to improve personnel satisfaction and avoid staff burnout (e.g. [Heimerl & Kolisch, 2010b](#); [Li et al., 2020](#); [Wang et al., 2022](#)). For example, [Gutjahr et al. \(2008\)](#) capture the learning and forgetting effect by defining competences of workers that can evolve over time, which influence the ability to work on future projects. They derive a worker efficiency value by applying a logistic transformation function and propose a nonlinear MIP formulation that they solve via a greedy heuristic and two meta-heuristics. [Mehmanchi and Shadrokh \(2013\)](#) consider learning and forgetting effects for the multi-skill resource-constrained project scheduling problem and aim to minimise the project makespan and maximise efficiency of workers simultaneously. However, no study considered learning effects to schedule staff more efficiently and realise more accurate staffing decisions.

The literature review pointed to the following shortcomings, i.e. (i) existing staffing models assume a static, predefined skill mix to determine the required budget, and (ii) the skill consequences of training have been neglected in associated personnel scheduling models. In the presented study, we propose a new staffing model and associated dedicated optimisation procedure incorporating the dynamic adaptation of the workers' skill mix during the planning horizon to determine the required budget more accurately. Accordingly, we consider the skill consequences of different types of training, learning and forgetting on the labour efficiency relative to the completion of regular tasks. Using this framework, insights are provided into the most cost-efficient mix for staffing and training in view of different input conditions concerning skill improvement rates, skill-efficiency transformation functions, and task set and workforce characteristics.

3. Gradual adaptation of skill level

To provide accurate staffing decisions and account for dynamic changes in the workforce skill level, a baseline schedule is composed. The underlying scheduling problem encompasses an integrated personnel days-off scheduling and task assignment problem, for which we assume that the demand and specific task characteristics (start times of tasks, duration of task modes, and staffing requirements) are known in advance. A worker's skill level is adapted when his/her skill level increases or decreases as a result of training, learning or forgetting. The prevailing skill level of a worker is calculated at the (fixed) moment tasks are initiated, impacting the duration for completing the tasks ([Gutjahr et al., 2008](#)). The scheduling of personnel resources given a fixed timing of activities has been considered in a wide range of studies, as in different application areas the task schedule and personnel schedule may be optimised in a sequential manner, i.e. the task schedule is composed before the personnel resource schedule is designed. This may be caused by the hierarchy of the different scheduling problems or the required timely communication of the task schedule to the stakeholders, amongst other reasons. Personnel resources are in most of those applications viewed as auxiliary resources to process the relevant tasks and are scheduled secondary. This assumption has also been hypothesised in the study of [Krishnamoorthy and Ernst \(2001\)](#), which

Table 1
Overview of learning and forgetting curves presented in the literature.

Model	Author(s)	Formula	Forgetting
Wright's Log-Linear model	Wright (1936)	$y = C_1 x^b$	
The Stanford-B model	Asher (1956)	$y = C_1(x + B)^b$	
DeJong's model	De Jong (1957)	$y = C_1(M + (1 - M)x^b)$	✓
The S-curve	Carr (1946)	$y = C_1(M + (1 - M)(x + B)^b)$	✓
The Plateau model	Baloff (1971)	$y = C + C_1 x^b$	
Two parameter hyperbolic learning curve	Mazur and Hastie (1978)	$y = k(\frac{x}{x+r})$	✓
Three parameter hyperbolic learning curve	Mazur and Hastie (1978)	$y = k(\frac{x+p}{x+p+r})$	✓
Knecht's exponential model	Knecht (1974)	$y = C_1 x^b e^{cx}$	
Three parameter exponential model	Mazur and Hastie (1978)	$y = k(1 - e^{-xp/r})$	
Time constant model	Towill (1990)	$y = y_c + y_f(1 - e^{-t/r})$	

proposed generic problem variants for the personnel task scheduling problem considering tasks having fixed start times.

In the following, we describe some example application areas that consider a fixed job schedule based on which the resource schedule (or staffing budget) needs to be devised. Many relevant studies relate to the production and project industry. Dondeti and Emmons (1992), amongst others, consider a task assignment problem in the production industry for which the processors differ in capacity and level of skills and the task schedule has been determined in advance. The goal is to determine the minimal cost relative to the devised resource schedules. Gutjahr et al. (2008) and Heimerl and Kolisch (2010a) indicate that project scheduling and staffing is traditionally decomposed into cascaded sub-problems which are solved separately. In the healthcare industry, when scheduling surgeries in the operating room department, the sequencing of the surgeries within each operating room is determined and surgery start times are fixed. The schedules for auxiliary resources such as medical residents and surgical nurses are built conforming to these fixed surgery schedules (Akbarzadeh et al., 2020; Brech et al., 2019). Lapègue et al. (2013) study an industrial problem that arises in companies specialised in drug evaluation and pharmacology research. Laboratories stipulate a very specific procedure, which contains a precise description of all the clinical tasks to be performed along with their fixed start times. Moreover, when composing resource schedules for many applications relative to the transportation sector, task schedules are also assumed to be fixed as they need to be communicated in advance to passengers or customers. Wang et al. (2023) study the scheduling of ground staff at airports, who perform various ground services such as passenger check in, package handling, cabin cleaning, and so on. The tasks for ground staff are determined in advance based on the flight schedule, which are considered for solving the staff rostering and task assignment problems.

As modelling associated skill adaptations tends to create a nonlinear model, we approximate the continuous learning curves via a discrete function, which is explained in Section 3.1. The formulas to gradually adapt the workers' competence scores resulting from learning and forgetting and each type of training can be found in Sections 3.2 and 3.3, respectively.

3.1. Discretisation of learning curves

Assume each task j of the set of tasks J (index j) corresponds to a task class $g \in G$. The binary parameter a_{jg} is equal to 1 if task j belongs to task class g and G is defined as the set of task classes, which group tasks that require the same set of skills. The adaptation of a worker's skill level may have a different effect for different task classes (Nembhard & Uzumeri, 2000b). For each task $j \in J$ and worker $w \in W$, we determine competence score o_{jw} at the fixed time s_j the task is initiated. This competence score expresses the skill level a worker possesses to complete tasks of a specific class, which is converted to an efficiency value that expresses the relation between

skill level and duration of a task. The duration of a task, carried out by a specific worker, is determined in two subsequent steps. First, the worker's competence score for the relevant skill is categorised in one of the discrete intervals with demarcated boundaries that match with a specific mode $e \in E$, with E defined as the set of possible task modes having a different task duration. Second, a transformation function is applied that converts the competence score into an efficiency value, which is used to determine the duration of the task.

The competences of a worker are quantified by the score o_{jw} , which represents the skill level of a worker w relative to task class g ($a_{jg} = 1$) at the moment task j is initiated. This score can be influenced in three ways, i.e. (i) the value o_{jw} increases if worker w effectuates training tasks that are targeted on improving skills relative to task class g ; (ii) the value o_{jw} increases if worker w carries out a (subset of) task(s) $j' \in A_j$, which consists of tasks relative to the same class g that are executed prior to task j , better known as learning; and (iii) the forgetting effect or so-called knowledge depreciation effect reduces the competence score in each period. The initial competence score for worker w relative to task class g is equal to i_{wg} . Based on a range of discrete values of competence scores o_{jw} , efficiency values h_{je} are derived to determine the appropriate task duration. To this end, we apply a nonlinear monotonous transformation function ϕ ensuring that the obtained values lie in the interval $[0, 1]$. An efficiency value of 0 means the task duration is equal to the baseline task duration, whereas an efficiency value of 1 implies the task duration is equal to the lowest possible duration according to which the task can be operated. To model the learning, we make use of a logistic function as proposed by Chen and Edgington (2005) and Ngwenyama et al. (2007). The applied transformation function can be found in Eq. (1) with parameters $a > 0$ and $b > 0$, specifying the input value domain.

$$\phi(o_{jw}) = \frac{1}{1 + ae^{-bo_{jw}}} \quad (1)$$

The purpose of this transformation function is to link competence scores to a particular efficiency value and corresponding task mode. Hence, the moment a task is initiated, we designate the task duration based on the interval that corresponds to the competence score of the worker. Each interval associated with a particular task mode is characterised by a minimum required competence score k_{je} and a maximum allowed competence score l_{je} . Eq. (2) models the mathematical relation between the efficiency value and the duration of a task mode (t_{je}). Each task class has a maximum percentage reduction r_g relative to the original task duration t_j as some tasks are more prone to be influenced by a higher skill level than others.

$$t_{je} = t_j - r_g h_{je} t_j \quad \forall j \in J, \forall e \in E_j \quad (2)$$

The concepts of skill adaptation described in further sections are illustrated via the example displayed in Table 2. We start by defining the intervals of the competence scores for the different modes by determining a minimum (k_{je}) and maximum (l_{je}) value. Efficiency values are determined based on the minimum competence score and

Table 2
Illustrative example defining multiple modes for a task.

Mode e	Minimum competence score k_{je}	Maximum competence score l_{je}	Efficiency value h_{je}	Duration t_{je}
1	$-\infty$	0.0499	0	432
2	0.05	0.0999	0.164	411
3	0.10	0.1499	0.326	390
4	0.15	0.1999	0.543	362
5	0.20	0.2499	0.745	335
6	0.25	0.2999	0.878	318
7	0.30	0.3499	0.947	309
8	0.35	0.3999	0.978	305
9	0.40	0.4499	0.991	304
10	0.45	0.4999	0.996	303
11	0.50	$+\infty$	0.998	303

the transformation function stated in Eq. (1). These efficiency values are used to set the duration of each mode based on the maximum possible reduction, which is in this example equal to 30% of the original task duration. Similar percentages have been observed in software development processes, for which the underlying problem encompasses a project scheduling problem (e.g. Guo et al., 2019). The composed table serves as input for the staffing problem under study. For each task j , we determine the competence score of a worker based on his/her initial competence score, the assigned regular and training tasks of the same task class, and the day at which task j starts. This score is categorised in one of the intervals and consequently the relevant modes and associated duration can be identified.

3.2. Learning and forgetting

In accordance with literature, we assume that the competence score of a worker increases with amount α , called the learning rate, each time the worker has performed a task. Similarly, we suppose that the competence score of a worker decreases each day with amount β , called the forgetting rate. We assume that the effect of learning is larger than the effect of forgetting ($\alpha > \beta$). The formula to calculate the competence score of a worker at the time task j is initiated, can be found in Eq. (3), which is based on research of Gutjahr et al. (2008). This competence score is the result of the initial competence of the worker for task class g relative to task j augmented with the product of the learning rate and the number of times a task of class g is executed by the worker prior to the start of task j , and decreased with the forgetting rate multiplied by day d_j on which task j needs to be executed.

$$o_{jw} = \sum_{g \in G} a_{jg} i_{wg} + \alpha \sum_{j' \in A_j} x_{j'w} - \beta d_j \quad \forall j \in J, \forall w \in W \quad (3)$$

In the example of Fig. 1, all three workers have an initial competence score of zero for all task classes. Two workers are able to reduce the processing times of tasks resulting from learning and forgetting, i.e. worker 1 for class 1 and worker 3 for class 4. The duration of task 15 is calculated as follows. Worker 1 has already executed two tasks of class 1, i.e. T1 and T8, with $\alpha = 0.10$ and task 15 takes place at day 2 (with $\beta = 0.05$) such that we obtain a competence score of 0.10 ($o_{jw} = 0 + 0.10 * 2 - 0.05 * 2$). This score corresponds to the interval associated with task mode 3 in Table 2. The corresponding efficiency value equals 0.326 and the task duration 390.

3.3. Training

In the following, we provide necessary context and equations to model the impact of on-the-job and off-the-job training on the competence score of a worker.

On-the-job training

We make distinction between two types of on-the-job training. First, a worker can follow an *internal training* where (s)he is assigned to explicit training tasks that involve e.g. instructions on specialised equipment to improve his/her skill level. These tasks, included in the set of training tasks T^o , are provided by an external trainer and can be planned very flexible because they are on request of the company and have a short duration. All on-the-job training tasks related to task class g of task j that are scheduled prior to task j are element of the set T_j^o and increase the skill level of a worker with improvement rate γ . Second, a worker w_2 can apprentice and learn from another company worker w_1 to execute task j in case both workers are assigned to this task, known as *shadow training*. In this way, worker w_2 can increase his/her skills for task class g ($a_{jg} = 1$) with improvement rate ϵ . As worker w_1 acts as tutor, (s)he needs sufficient knowledge, having a competence score that is at least p higher than the competence score of coworker w_2 .

Eq. (4) determines the competence score of a worker at the start of task j as a result of on-the-job training. Similar to the learning (cf. Section 3.2), a worker's initial competence score is increased with the number of training tasks of class g that are executed by the worker prior to task j , multiplied by the skill improvement rate for internal training. We further augment the competence score with the product of the learning rate for shadow training and the number of times worker w has been an apprentice for a task $j' \in A_j$. Gathered skills acquired via training could be partly forgotten by the worker, which implies a daily reduction of the competence score with the forgetting rate.

$$o_{jw} = \sum_{g \in G} a_{jg} i_{wg} + \gamma \sum_{j' \in T_j^o} x_{j'w} + \epsilon \sum_{j' \in A_j} q_{j'w} - \beta d_j \quad \forall j \in J, \forall w \in W \quad (4)$$

with

- q_{jw} 1, if worker $w \in W$ is apprentice for task $j \in J$; and 0, otherwise
- A_j Set of tasks that start earlier than task j and have the same task class

In Fig. 2, an example of both shadow training and internal training is provided. Worker 3 is assigned to training task *Tr1* on day 1, given by an external trainer for task class 3. Note that the timing of this training task is very flexible, i.e. internal training tasks can start every hour so that *Tr1* is scheduled between task 7 and task 10. By following this internal training, the competence score of worker 3 is increased by γ ($= 0.20$). In this way, worker 3 can process tasks of class 3 in a smaller duration. The competence score o_{jw} for worker 3 at the start of task 10 is equal to 0.15 ($0 + 0.20 + 0 - 0.05$) such that the task can be processed following task mode 4 (duration: 362 min) instead of task mode 1 (duration: 432 min). Furthermore, the concept of shadow training is illustrated with regards to task 10. Worker 1 is an apprentice for this task with worker 3 as tutor. Worker 3 has a competence score that is more than p higher than worker 1, who is apprentice and increases his/her competence score with ϵ ($= 0.15$). In this way, the competence score of worker 1 equals 0.05 ($0 + 0 + 0.15 - 0.05 * 2$) on day 2, such that (s)he may carry out task 11 according to task mode 2 having a lower duration equal to 411.

Off-the-job training

Off-the-job training may comprise different variants, depending on the goal of the training. The variants may assume a one-day training (e.g. conference, lecture, vestibule training), a half-a-day training (e.g. sensitivity and transactional training) or a quarter-of-a-day training (e.g. simulation exercises) and increase the competence score of worker with improvement rate χ , ψ or ω , respectively. Each of these training variants are taught via particular training tasks of respective sets T^{f1} , T^{f2} and T^{f3} . The timing of this external training is fixed, which entangles the fitting of these training tasks in the lines-of-work

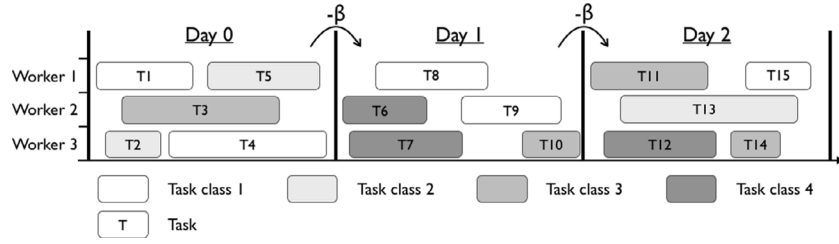


Fig. 1. Example illustrating learning and forgetting.

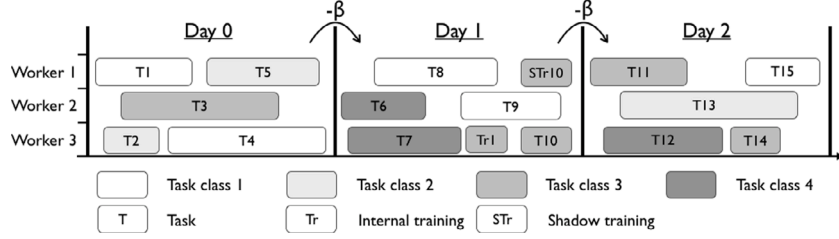


Fig. 2. Example illustrating on-the-job training.

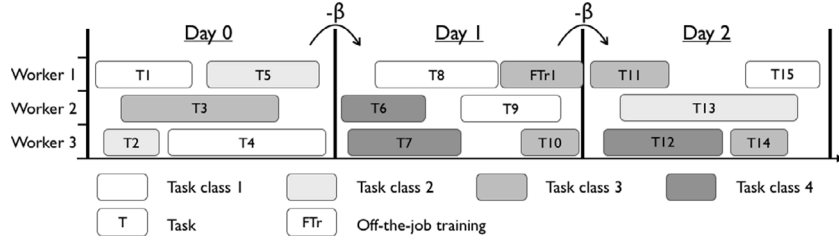


Fig. 3. Example illustrating off-the-job training.

of workers. The competence score at the start of task j resulting from this type of training is calculated via Eq. (5). The initial competence of a worker is increased with the improvement rate of the training multiplied by the number of times a training task of class g is assigned to the worker prior to the start of task j . Further, we account for the forgetting effect, decreasing the competence score with the product of the forgetting rate and the day task j is carried out.

$$o_{jw} = \sum_{g \in G} a_{jg} i_{wg} + \chi \sum_{j' \in T_j^{f1}} x_{j'w} + \psi \sum_{j' \in T_j^{f2}} x_{j'w} + \omega \sum_{j' \in T_j^{f3}} x_{j'w} - \beta d_j \quad \forall j \in J, \forall w \in W \quad (5)$$

Off-the-job training is illustrated in Fig. 3. In the example, worker 1 is able to attend a training of task class 3 that takes a quarter of a day on day 1, increasing his/her competence score with ω ($= 0.35$). Thanks to this training, the worker's competence score is equal to 0.25 ($0 + 0 + 0 + 0.35 - 0.05 * 2$) at the time task 11 is initiated on day 2. The duration of task 11 may be reduced from 432 to 318 min. This way, more tasks may be included in the same line-of-work, potentially reducing the number of required resources.

4. The integrated days-off scheduling and personnel task assignment problem

In order to construct the staffing budget accurately, we compose a baseline schedule by solving the integrated days-off scheduling and personnel task assignment problem. This section details this underlying scheduling problem with possibility to adapt the workers' skill mix throughout the planning horizon. The baseline scheduling problem

assigns workers to duties and to individual tasks, composing a line-of-work, such that the staffing requirement R_j of each task is satisfied. We assume a set of workers W (index w) and a set of task classes G (index g). A worker w can be hired at cost c_w , which depends on his/her initial competence i_{wg} . The skill level of workers, however, can change over time as the execution of the schedule gradually progresses. Based on the skill level of the worker at the time a task is initiated, the duration of task j and corresponding task mode can be selected out of a set of discrete modes E_j . The baseline personnel roster is created for a certain planning horizon consisting of the set of days D (index d). By introducing different task classes, task complexity and the nature of the task can be incorporated. Each task class $g \in G$ has similar characteristics such as cognitive, complex motor tasks, etc, which are impacted in the same manner by a change in skill level. Further, each task is represented by a fixed start time s_j and a variable finishing time f_{je} depending on the chosen mode, which determines the task duration. The start and finish times of the modes are used to construct a set of maximal cliques C (index c). Each clique consists of a set of task modes K_c that overlap for a certain time interval so that appropriate scheduling avoids that two overlapping tasks are to be completed by the same worker. Individual workers are scheduled over the horizon by assigning them to days on and days off, taking time-related constraints into account. A line-of-work is defined as an individual duty roster for a worker, which assigns days on to the worker during which (s)he executes a number of tasks. Mathematical formulations considering training, learning and forgetting are stated in following sections.

4.1. Learning and forgetting

In this section, we provide the mathematical formulation of the integrated days-off scheduling and personnel task assignment problem with

learning and forgetting. To correctly calculate the worker competence score at the time tasks are initiated, Eq. (3) is considered.

List of symbols

Sets

W	Set of personnel (index w)
D	Set of days in the planning horizon (index d)
N	Set of weeks in the planning horizon (index n)
J	Set of tasks (index j)
G	Set of task classes (index g)
E_j	Set of modes relative to task j (index e)
C	Set of maximal cliques (index c)
K_c	Set of overlapping task modes incorporated in the maximal clique c
A_j	Set of tasks that start earlier than task j and have the same task class

Parameters

a_{jg}	1, if task j belongs to task class g ; 0, otherwise
R_j	Staffing requirement of task j
d_j	Day on which task j needs to be executed
s_j	Start time of task j
f_{je}	Finish time of mode e , task j
k_{je}	Minimum required competence score to perform mode e of task j
m	Maximum number of working days in one week for a single worker
c_w	Hiring cost of worker w
M	Big number
α	Learning rate
β	Forgetting rate

Decision variables

x_{jw}	1, if task j is assigned to worker w ; 0, otherwise
y_w	1, if worker w is assigned to a line-of-work with actual work days; 0, otherwise
z_{je}	1, if mode e is selected for task j ; 0, otherwise
v_{wd}	1, if worker w has to work on day d ; 0, otherwise
o_{jw}	Competence score of worker w at the time task j is initiated
a_{wje}^1	1, if worker w satisfies the minimum competence score to perform mode e of task j ; 0, otherwise

Mathematical formulation

$$\text{Min } Z = \sum_{w \in W} c_w y_w \quad (6)$$

s.t. Constraint (3)

$$\sum_{w \in W} x_{jw} = R_j \quad \forall j \in J \quad (7)$$

$$\sum_{(j,e) \in K_c} z_{je} x_{jw} \leq y_w \quad \forall w \in W, \forall c \in C \quad (8)$$

$$\sum_{e \in E_j} z_{je} = 1 \quad \forall j \in J \quad (9)$$

$$\sum_{j \in J | d_j = d} x_{jw} \leq M v_{wd} \quad \forall w \in W, \forall d \in D \quad (10)$$

$$\sum_{d=7n}^{7n+6} v_{wd} \leq m \quad \forall w \in W, \forall n \in N \quad (11)$$

$$o_{jw} \leq k_{je} + M a_{wje}^1 \quad \forall w \in W, \forall j \in J, \forall e \in E_j \quad (12)$$

$$o_{jw} \geq k_{je} - M(1 - a_{wje}^1) \quad \forall w \in W, \forall j \in J, \forall e \in E_j \quad (13)$$

$$z_{je} x_{jw} \leq a_{wje}^1 \quad \forall w \in W, \forall j \in J, \forall e \in E_j \quad (14)$$

$$x_{jw} \in \{0, 1\} \quad \forall w \in W, \forall j \in J$$

$$y_w \in \{0, 1\} \quad \forall w \in W$$

$$\begin{aligned} v_{wd} &\in \{0, 1\} & \forall w \in W, \forall d \in D \\ o_{jw} &\geq 0 & \forall j \in J, \forall w \in W \\ z_{je} &\in \{0, 1\} & \forall j \in J, \forall e \in E_j \\ a_{wje}^1 &\in \{0, 1\} & \forall w \in W, \forall j \in J, \forall e \in E_j \end{aligned} \quad (15)$$

The objective function (6) minimises the hiring cost required to cover all tasks. Eq. (3) encompasses the calculation of the competence score o_{jw} of an employee at the fixed moment a task is initiated based on his/her initial competence, learning and forgetting. Via this score, a mode can be selected for completing the task aligned with the acquired competence of the allocated worker(s). Eq. (7) ensures that every task is executed by the number of workers postulated by the staffing requirement R_j . Constraint (8) models the relation between tasks, modes and workers, and ensures that whenever a worker is assigned to a task (mode), the worker is hired. Moreover, this constraint imposes that only one mode in a clique can be assigned to avoid overlap between task modes assigned to a worker. The requirement of selecting exactly one mode for each task is stated in Eq. (9). Hence, in case multiple workers are required to process a task, the task is executed following a task efficiency stipulated by the worker with the lowest skill level. Constraint (10) makes sure that once a worker is assigned to one or more tasks, the associated day is designated as a working day for that worker. Eq. (11) restricts the number of working days per week for each worker. Constraints (12) and (13) determine if a worker has acquired the minimum competence required to execute a task according to a specific task mode, which is modelled via an either-or constraint structure. The left-hand sides of these constraints equal the competence score of a worker at the moment a task is initiated, which is compared to the minimum required competence at the right-hand sides. When the worker meets the minimum competence score, constraint (13) is relevant and variable a_{wje}^1 is equal to 1, whereas constraint (12) is redundant via the big-M value. If not, variable a_{wje}^1 equals 0 and constraint (12) is relevant. If a worker does not possess the minimum competence requirements at the time the task is initiated, Eq. (14) imposes that (s)he cannot be assigned to the associated task mode. These constraints allow a worker may complete a task following a mode having a longer duration than designated by his/her competence score. Eq. (15) presents domain constraints for all variables involved.

4.2. Training

In this section, we discuss the required adaptation of the model presented in Section 4.1 to the scheduling problem under study for different forms of training.

On-the-job training

To solve the scheduling problem with inclusion of shadow training and/or internal training, we consider the variant model (16)–(23) with constraints (4) and (8)–(15) appended. Note that in this model, the set of maximal cliques C is expanded with the set of internal on-the-job training activities T^o and we iterate over both sets J and T^o in constraint (10).

Additional symbols

Sets

T_j^o	Set of on-the-job internal training tasks starting prior to task j with the same task class
---------	---

Parameters

p	Minimum competence difference between tutor and apprentice
γ	Skill improvement rate of internal training
ϵ	Skill improvement rate of shadow training

Decision variables

q_{jw}	1, if worker w is apprentice for task j ; 0, otherwise
$b_{jw'w}^1$	1, if worker w can act as a tutor to worker w' for task j ; 0, otherwise

$$\text{Min } Z = M \sum_{w \in W} c_w y_w + \sum_{j \in T^0} \sum_{w \in W} x_{jw} + \sum_{j \in J} \sum_{w \in W} q_{jw} / M \quad (16)$$

s.t. Constraints (4), (8)–(15)

$$\sum_{w \in W} x_{jw} = R_j + \sum_{w \in W} q_{jw} \quad \forall j \in J \quad (17)$$

$$z_{j0} \geq x_{jw} \quad \forall j \in T^0, \forall w \in W \quad (18)$$

$$o_{jw} - o_{jw'} \geq p - M(1 - b_{jw'w}^1) \quad \forall j \in J, \forall w, w' \in W \quad (19)$$

$$x_{jw} \geq 1 - M(1 - b_{jw'w}^1) \quad \forall j \in J, \forall w, w' \in W \quad (20)$$

$$x_{jw'} \geq 1 - M(1 - b_{jw'w}^1) \quad \forall j \in J, \forall w, w' \in W \quad (21)$$

$$q_{jw'} \leq \sum_{w \in W} b_{jw'w}^1 \quad \forall j \in J, \forall w' \in W \quad (22)$$

$$q_{jw}, b_{jw'w}^1 \in \{0, 1\} \quad \forall j \in J, \forall w, w' \in W \quad (23)$$

The objective function (16) minimises primarily the staffing cost. However, we also minimise the number of internal and shadow training sessions. As internal training possibly causes a training cost due to the hiring of an external trainer whereas shadow training is given by one of the company workers, the weight for shadow training is small ($1/M$) such that objectives are optimised according to different priorities. Eq. (4) calculates the competence scores of workers resulting from on-the-job training and forgetting at the time a task is initiated. Eq. (17) stipulates the staffing requirements. This constraint enables the modelling of workers as apprentice workers when more resources are assigned to a task. Constraint (18) models the assignment of a worker to an internal training task and ensures that if a worker is assigned to such a training task ($x_{jw} = 1$), the single task mode, i.e. mode 0, is selected ($z_{j0} = 1$). Note that these training tasks are taken into account in the computation of the maximal cliques, such that there is no overlap between training and other tasks (cf. Eq. (8)). Shadow training is only feasible if the following conditions are satisfied, i.e. (i) constraint (19) imposes that the competence difference between tutor and apprentice should be sufficiently large; and (ii) both the tutor and the apprentice should be assigned to the same task, which is modelled via the indicator variable $b_{jw'w}^1$ in constraints (20) and (21), respectively. Constraint (22) evaluates for each task if there are two workers, i.e. a tutor and an apprentice, who satisfy the necessary conditions by iterating over the auxiliary variables $b_{jw'w}^1$, and sets the task-apprentice variable $q_{jw'}$ accordingly to correctly calculate the skill level of workers. The domain constraints of the task apprentice and auxiliary variables can be found in Eq. (23).

Off-the-job training

The model involving off-the-job-training, is very similar to the model presented in Section 4.1. More precisely, we consider constraints (7)–(15) and Eq. (5) to calculate adjustments to the competence scores correctly. Constraint (25) is appended relative to all types of external training, which is similar to constraint (18). The main change to the mathematical formulation is the objective function (24). This objective minimises primarily the staffing cost, but also minimises training costs associated with the three different types of external training given respective unit costs c^{f1} , c^{f2} and c^{f3} . To compose the model for off-the-job training correctly, constraints (8) and (10) consider the external training tasks.

Additional symbols

Sets

- T_j^{f1} Set of one-day off-the-job training tasks starting prior to task j with the same task class
- T_j^{f2} Set of half-a-day off-the-job training tasks starting prior to task j with the same task class
- T_j^{f3} Set of quarter-of-a-day off-the-job training tasks starting prior to task j with the same task class

Parameters

- χ Skill improvement rate of one-day off-the-job training
- ψ Skill improvement rate of half-a-day off-the-job training
- ω Skill improvement rate of quarter-of-a-day off-the-job training

$$\begin{aligned} \text{Min } Z = M \sum_{w \in W} c_w y_w + c^{f1} \sum_{j \in T^{f1}} \sum_{w \in W} x_{jw} \\ + c^{f2} \sum_{j \in T^{f2}} \sum_{w \in W} x_{jw} + c^{f3} \sum_{j \in T^{f3}} \sum_{w \in W} x_{jw} \end{aligned} \quad (24)$$

s.t. Constraints (5), (7)–(15)

$$z_{j0} \geq x_{jw} \quad \forall j \in T_j^{f2} \cup T_j^{f1} \cup T_j^{f3}, \forall w \in W \quad (25)$$

5. Solution procedure

In this section, we propose a dedicated procedure for solving the problem under study stated in Section 4 where tasks have a predefined resource requirement; i.e. the procedure is applicable for the integrated days-off scheduling and personnel task assignment problem considering learning and forgetting and/or off-the-job training. Accordingly, the proposed decomposition is not compatible for solving the problem considering shadow training, because this type of on-the-job training requires the dual assignment of an apprentice and a tutor to the same task. To benchmark this form of training to others, we rely on solving the proposed MIP model (16)–(23) via mathematical programming. In the following, we discuss the solution procedure for the problem considering traditional learning but an identical optimisation search is conducted when considering off-the-job training. This procedure is proposed because using a commercial solver to solve this problem is detrimental for the computational performance, due to the large amount of symmetries that are present resulting from the objective function structure. Therefore, we propose a matheuristic procedure that combines the fixing of lines-of-work for some employees based on a high-quality solution derived by a constructive heuristic and a branch-and-price procedure that optimises the coverage of the tasks that have not been assigned yet.

Algorithm 1 Overview procedure

- 1: Determine lower bound using maximum efficiency levels (Section 5.1) *Step 1*
- 2: Compose upper bound via constructive heuristic (Section 5.2) *Step 2*
- 3: **if** (upper bound > lower bound) **then** *Step 3*
- 4: Fix lines-of-work for a subset O of the assigned workers (Section 5.3)
- 5: Perform constructive heuristic for tasks $j \in J \mid \sum_{w \in O} x_{jw} < R_j$ (Section 5.2)
- 6: **if** (upper bound > lower bound) **then** *Step 4*
- 7: Solve branch-and-price algorithm for tasks $j \in J \mid \sum_{w \in O} x_{jw} < R_j$ (Section 5.4)

Algorithm 1 shows the different steps of the proposed procedure. In *Step 1*, a dedicated lower bound is calculated assuming that all workers operate according to the maximum efficiency level. In *Step 2*, we solve the problem taking advancing competence scores into account using a constructive heuristic that solves the longest path problem until at least $|J| - Q$ of the $|J|$ tasks are covered. The remainder of the tasks that are not covered, are staffed by solving the relevant model via a commercial solver. This initial solution comprises an upper bound that is input to the branch-and-price procedure. When this solution is equal to the lower bound, the algorithm can be stopped and an optimal solution has been found. Otherwise, we invoke *Step 3* that fixes lines-of-work and associated task-worker assignments following the initial solution for a subset $O \subseteq W$ of the workers hired. The size of the subset O is dependent on the instance size. Afterwards, the constructive heuristic is repeated for those tasks that are not covered completely by the fixed

lines-of-work ($j \in J \mid \sum_{w \in O} x_{jw} < R_j$). When this new solution improves the solution found so far, the upper bound is updated and again the optimality of the solution is checked by comparing lower and upper bound solutions. If the current solution is not optimal, a branch-and-price algorithm is invoked in *Step 4* to solve the problem for all tasks $j \in J \mid \sum_{w \in O} x_{jw} < R_j$.

5.1. Lower bound

To reduce the run time of the proposed procedure significantly, we try to construct a tight lower bound. On the one hand, a good lower bound can close the optimality gap such that the algorithm can be terminated earlier after *Step 2*. On the other hand, the lower bound could stop the CG procedure earlier and/or reduce the size of the branching tree. To define an appropriate lower bound, we simplify the mathematical formulation of Section 4 by excluding constraints (12)–(14) and introducing constraint (26). This latter equation assumes each worker operates according to the highest competence score, such that each task can be executed in its shortest possible duration, leading to a lower bound on the hiring cost required to cover all tasks.

$$z_{je} = 1 \quad \forall j \in J, \forall e \in E_j : l_{je} = +\infty \quad (26)$$

5.2. Constructive heuristic

The heuristic presented in Algorithm 2 constructs an initial schedule by solving a so-called longest-path problem per worker that tries to include as many tasks as possible in a line-of-work. The heuristic starts with putting the set of uncovered tasks $\hat{J}$ equal to J , the staffing cost $\hat{Z}$ relative to the workers hired equal to 0, the uncovered staffing requirements $\hat{R}_j$ equal to R_j ($\forall j \in J$), and the set of ineligible modes for each task $\hat{E}_j$ as empty. Furthermore, the workers are sorted in ascending order of hiring cost c_w . In every iteration, a line-of-work is determined for a particular worker by maximising the number of tasks that can be staffed. Based on this line-of-work, the temporary staffing requirements $\hat{R}_j$ and the sets of ineligible modes $\hat{E}_j$ of the covered tasks are updated. The ineligible modes of a covered task involves (i) modes for which the worker does not possess the required minimum competence score at the time the task is initiated and (ii) modes that could cause overlap with the consecutive tasks allocated to the same line-of-work. Tasks that have reached their required number of workers ($\hat{R}_j = 0$) are removed from the set of uncovered tasks $\hat{J}$. Each time a line-of-work is composed, the staffing cost $\hat{Z}$ is increased by c_w . This procedure is repeated until the number of uncovered tasks is lower than a predefined number Q .

Algorithm 2 Overview constructive heuristic

```

1: Initialise  $\hat{J} = J$ ,  $\hat{Z} = 0$ ,  $\hat{R}_j = R_j$  ( $\forall j \in J$ ) and  $\hat{E}_j = \emptyset$  ( $\forall j \in J$ )
2: do
3:   Solve longest path problem
4:   for each task  $j \in \hat{J} \mid x_j = 1$  do
5:     Lower uncovered staffing requirement by one, i.e.  $\hat{R}_j = \hat{R}_j - 1$ 
6:     Update  $\hat{E}_j = \{e \in E_j \mid (o_{jw} < k_{je}) \vee (\exists j' \in \hat{J} \setminus j : x_{j'w} = 1 \wedge f_{je} \in [s_{j'}, f_{j'}]_{E_j})\}$ 
7:   if ( $\hat{R}_j = 0$ ) then
8:     Remove task  $j$  from  $\hat{J}$ 
9:   Increase number of hired workers and corresponding staffing cost, i.e.  $\hat{Z} = \hat{Z} + c_w$ 
10: while  $|\hat{J}| > Q$ 
11: Solve model (6)–(15) for  $\hat{J}$  and increase  $\hat{Z}$  by  $Z$ 
12: if ( $\hat{Z} < Z^{inc}$ ) then
13:   Set  $Z^{inc} = \hat{Z}$ 
14: for each task  $j \in \hat{J}$  do
15:   Increase  $c_j$  with  $M$ 
16: if ( $Z^{inc} > \text{lower bound}$ ) then
17:   Repeat line 1–13 with adjusted objective function coefficients  $c_j$ 

```

To determine the longest line-of-work for an individual worker, we rely on mathematical programming and solve model (27)–(31) for worker w to optimality. The model includes constraints (10)–Eq. (14) relative to a single worker. In addition, depending on whether the worker's skill level is adapted via learning or training, Eqs. (3), (4) or (5) needs to be included. Eq. (27) maximises the reward of the tasks included in the longest path and associated the number of tasks ($c_j > 0$). Each task has a corresponding reward c_j , which is initially the same for all tasks ($c_j = 1$). To prevent that overlapping tasks are selected, constraint (28) is included. Eq. (29) assures that if a task is included, a corresponding mode is selected. Eq. (30) prevents the selection of an ineligible mode. Constraint (31) defines the domain for variables x_{jw} and z_{je} .

$$\text{Max } \hat{Z} = \sum_{j \in J} c_j x_{jw} \quad (27)$$

s.t. Constraints (3) or (4) or (5), (10)–(14)

$$\sum_{(j,e) \in K_c} z_{je} x_{jw} \leq 1 \quad \forall c \in \hat{C} \quad (28)$$

$$\sum_{e \in E_j} z_{je} = x_{jw} \quad \forall j \in \hat{J} \quad (29)$$

$$z_{je} = 0 \quad \forall j \in \hat{J}, \forall e \in \hat{E}_j \quad (30)$$

$$x_{jw}, z_{je} \in \{0, 1\} \quad \forall j \in J, \forall e \in E_j \quad (31)$$

with

c_j Reward for including task $j \in \hat{J}$ in the longest path
 $\hat{C}$ Set of maximal cliques relative to set of tasks $\hat{J}$ (index c)

From the moment fewer than Q tasks need to be covered, we determine for those remaining tasks the minimum required hiring cost by solving model (6)–(15) via mathematical programming, because the constructive heuristic will rapidly increase the number of required employees when tasks cannot be easily combined. The resulting hiring cost Z is added to the cost $\hat{Z}$ of currently hired workers. If this aggregated cost is lower than the best solution found so far, the incumbent solution Z^{inc} is updated. If the yielded upper bound is larger than the lower bound (cf. Section 5.1), the constructive heuristic is repeated with prioritised objective coefficients for those tasks that are the hardest to include in a line-of-work. More precisely, we increase the reward c_j by a priority coefficient M for the tasks $j \in \hat{J}$ that were not included in one of the lines-of-work at the moment fewer than Q tasks were needed to be covered. In this way, prioritised tasks will be rapidly included in a line-of-work and a different heuristic solution will be constructed.

5.3. Line-of-work fixing

In order to accelerate the search for a high-quality integer solution, we simplify the problem by fixing a predefined number of lines-of-work $|O|$, reducing the large amount of symmetries in the solution space. Parameter $|O|$ is defined by the user. The height of this parameter determines the required computational effort of later optimisation steps. When $|O| = 0$, no lines-of-work are fixed, and all assignments are derived following further optimisation. When $|O| = |W|$, the schedule that has been generated via the constructive heuristic is fixed. We suggest adapting the number of fixed lines-of-work to the instance size such that a good balance can be found between solution quality and computational performance. As the constructive heuristic composes two solutions, i.e. one based on the rewards $c_j = 1$ and one based on the prioritised rewards $c_j \neq 1$, we select the best solution with the lowest hiring cost $\hat{Z}$. Selection of lines-of-work that are fixed is based on a decreasing order relative to the number of tasks scheduled, i.e. we

fix the lines-of-work with a larger number of tasks. As these lines-of-work are fixed, we do not solve the associated pricing problem for the involved workers following the branch-and-price procedure in further calculations. In addition, when solving the master problem, some of the task modes may be ineligible for other workers due to the composition of the fixed lines-of-work (cf. Section 5.2). The set of ineligible modes for task j is denoted by $\hat{E}_j$.

5.4. Branch-and-price

The remaining tasks are staffed via a dedicated branch-and-price procedure. To this end, the underlying integrated personnel days-off scheduling and task assignment problem is reformulated based on the Dantzig–Wolfe decomposition, which leads to the definition of a master and subproblem (Dantzig & Wolfe, 1960). The variables of the master problem represent lines-of-work that encompass a sequence of working days and days off for an individual employee, and tasks assignments that are completed by the worker on every working day taken his/her efficiency level at the time of task execution into account. The formulation of the master problem is discussed in detail in Section 5.4.1. As there is a huge amount of individual lines-of-work possible, the standard solution approach is CG to solve the linear programming (LP) relaxation of the master problem efficiently (Barnhart et al., 1998). A CG procedure considers only a limited number of potential lines-of-work in the so-called restricted master problem (RMP) and relies on a multi-stage pricing procedure discussed in Section 5.4.2 to identify promising columns with negative reduced cost. The LP lower bound relative to each CG iteration is tightened by rounding the LP solution up to the closest integer. When the pricing procedure is able to find promising columns, these are added to the master problem and a new CG iteration is started. Once no more columns can be added or the LP solution value is lower than the lower bound, an optimal LP solution has been found for the master problem and the CG procedure can be stopped. However, as solving the LP relaxation could lead to fractional solutions, branching is applied when the retrieved LP solution does not satisfy the integer domain conditions. Branching divides larger problems into smaller non-overlapping problems, represented by nodes, to search for an integer optimal solution. In Section 5.4.3, a variant of follow-on branching is proposed and a depth-first strategy is applied that selects the next unexplored node. A CG procedure is invoked in every node of the branching tree considering appropriate branching constraints until an optimal integer solution has been found.

5.4.1. Master problem

Every iteration of the CG algorithm, an RMP is solved that assigns workers to a line-of-work. As including all possible lines-of-work is not feasible, only a restricted set of columns L (index l) is considered that comprises the task (mode) assignments and the days-off pattern. Note that the column l_0 represents an empty line-of-work enabling a proper staffing decision and that each worker can only be assigned to a subset of columns $L_w \subseteq L$ based on his/her initial competences. The mathematical formulation of the RMP is provided below.

List of additional symbols

Sets

L_w Set of lines-of-work for worker w (index l)

Parameters

δ_{lj} 1, if task j is covered by column l ; 0, otherwise

η_{lje} 1, if mode e of task j is used in column l ; 0, otherwise

Decision variables

u_{wl} 1, if line-of-work l is assigned to worker w ; 0, otherwise

Mathematical formulation

$$\text{Min } Z' = \sum_{w \in W} \sum_{l \in L_w \setminus l_0} c_w u_{wl} \quad (32)$$

$$\text{s.t. } \sum_{w \in W} \sum_{l \in L_w} \delta_{lj} u_{wl} = R_j \quad \forall j \in J \quad (\kappa_j) \quad (33)$$

$$\sum_{l \in L_w} u_{wl} = 1 \quad \forall w \in W \quad (\lambda_w) \quad (34)$$

$$\sum_{e \in E_j \setminus \hat{E}_j} z_{je} = 1 \quad \forall j \in J \quad (\mu_j) \quad (35)$$

$$\sum_{l \in L_w} \eta_{lje} u_{wl} \leq z_{je} \quad \forall j \in J, \forall e \in E_j \setminus \hat{E}_j, \forall w \in W \quad (\pi_{jew}) \quad (36)$$

$$z_{je} \in \{0, 1\} \quad \forall j \in J, \forall e \in E_j \setminus \hat{E}_j$$

$$u_{wl} \in \{0, 1\} \quad \forall w \in W, \forall l \in L \quad (37)$$

The objective (32) of the RMP is to minimise the worker hiring cost for covering all tasks. Eq. (33) embodies the staffing requirement for each task. Each worker can only be assigned to a single line-of-work as stated in Eq. (34). Eq. (35) stipulates that only one mode can be chosen for each task, while constraint (36) ensures that the task mode variables are aligned with the selected modes in the lines-of-work. Constraints (37) indicate the domains of the task mode and column variables.

5.4.2. Pricing problem

Assume κ_j , λ_w , μ_j and π_{jew} be the respective dual prices associated with constraints (33), (34), (35) and (36). The dual prices associated with an optimal solution of the RMP are used to determine promising columns that could possibly improve the LP solution value. The pricing problem is solved only once per subset of workers having the same initial competence score for each skill class (i_{wg}). Furthermore, separate pricing problems are solved for workers whose assignments are restricted due to earlier imposed branching constraints. Note that the pricing problem is not solved for workers whose line-of-work has been fixed (cf. Section 5.3). The objective function (38) of the pricing problem for worker w minimises the reduced cost. Further, the pricing problem considers constraints (8)–(15), relative to the scheduling of a single worker. Note that to determine the reduced cost for workers for who no branching constraints have been imposed earlier, we consider the dual price λ_w with the largest value over all workers having the same initial competence score. These columns are added to the RMP for every relevant worker.

$$\text{Min } Z'' = c_w - \sum_{j \in J} \kappa_j x_{jw} - \lambda_w - \sum_{j \in J} \mu_j x_{jw} - \sum_{j \in J} \sum_{e \in E_j} \pi_{jew} z_{je} \quad (38)$$

Multi-stage pricing solution methodology

As it is computationally extensive to determine in each iteration the column with the most negative reduced cost, it is sufficient to determine one (or multiple) column(s) with negative reduced cost. Therefore, we propose a multi-stage pricing procedure that makes a trade-off between the potential improvement that is obtained in the LP solution value by adding one or more new columns and the computational effort, solving the pricing problem exactly or heuristically. The pricing procedure that is used to find new columns for workers having initial competence score i_{wg} can be found in Algorithm 3.

To determine whether we solve the pricing problem using an exact and/or heuristic procedure, we evaluate two conditions, i.e. (i) we count the number of consecutive iterations (H^{iter}) the pricing problem is solved heuristically and compare this number with the maximum allowed number (H^{max}) and (ii) we evaluate if the potential improvement relative to the LP solution value is lower than a user-defined threshold value LP^{imp} . For the latter, we compare the LP solution value of the RMP obtained in the previous CG iteration (Z'^{prev}) with the LP solution value of the current iteration ($Z'^{current}$). Note that for the first CG

iteration in any node, Z'^{prev} is equal to $+\infty$. If this difference is smaller than LP^{imp} , this may hint to the fact that the potential improvement the heuristic may offer is too limited. If one of these two conditions is satisfied, we invoke *Stage 1* and solve the pricing problem, defined via Eqs. (8)–(15) and (38), exactly via mathematical programming to determine the column with the most negative reduced cost. Further in this stage, we solve the pricing problem heuristically to derive multiple diverse patterns. As the problem has been solved exactly in this stage, H^{iter} is reinitialised to zero. However, if none of these conditions are satisfied, we invoke *Stage 2*. Because always solving the pricing problem using mathematical programming is computationally not efficient, we only apply a heuristic procedure, increasing H^{iter} by one. Only when this procedure is not able to find any promising column, the pricing model is solved exactly for the selected worker. If no column can be found with negative reduced cost in Stage 1 or 2 for workers without having branching conditions imposed, the pricing problem is solved exactly in *Stage 3* for the workers with branching conditions imposed to identify promising columns.

Algorithm 3 Multi-stage pricing procedure (i_{wg})

```

1: Determine worker  $w$  with competence  $i_{wg}$  having the highest dual price  $\lambda_w$ 
2: if  $((H^{iter} = H^{max}) \text{ or } (Z'^{prev} - Z'^{current} < LP^{imp}))$  then Stage 1
3:   Solve pricing problem for worker  $w$  using mathematical programming
4:   Solve pricing problem for worker  $w$  using Heuristic pricing procedure
5:   Set  $H^{iter} = 0$ 
6: else Stage 2
7:   Solve pricing problem for worker  $w$  using Heuristic pricing procedure
8:   Increase  $H^{iter}$  by one
9:   if (No column found with negative reduced cost) then
10:    Solve pricing problem for worker  $w$  using mathematical programming
11: if (No column found with negative reduced cost) then Stage 3
12:   Solve pricing problem for all workers with competence  $i_{wg}$  having branching constraints imposed using mathematical programming

```

Heuristic pricing procedure

The heuristic pricing procedure as stated in Algorithm 4 tries to find a column with negative reduced cost using two different search methods, i.e.

- A *single-pass heuristic* that produces U lines-of-work for each possible days-off pattern with the maximum amount of working days. For each days-off pattern, we iterate over all tasks related to one of the stipulated working days and define a Bernoulli distribution with a probability of selection for each task based on its observed dual price κ_j . A random number is generated to determine if a task is to be included in a line-of-work or not on the condition that there is no overlap with already scheduled tasks for the worker. If a task is included, we update the competence score of the worker. After constructing columns for the different days-off patterns, we add all columns with negative reduced cost to the master problem. Two different approaches are used to include tasks in a line-of-work, i.e. (i) a probability $P^1 > 0$ is defined for tasks with $\kappa_j > 0$, while tasks that have a dual price ≤ 0 get a (lower) probability $P^2 > 0$; and (ii) when the previous approach has not lead to any promising column, a new line-of-work is composed by potentially including only those tasks with probability P^1 that have the potential to reduce the cost of the column, i.e. with $\kappa_j > 0$. The selection probability P^2 for tasks with $\kappa_j \leq 0$ is equal to 0. Note that for both approaches, tasks are processed in a logical order based on start time.

Algorithm 4 Heuristic pricing procedure

```

1: for each possible days-off pattern with  $m$  working days do Single-pass heuristic
2:   counter = 0
3:   do
4:     for each task  $j \in J$  with  $d_j$  being a working day in the selected days-off pattern do
5:       if (task  $j$  has no overlap with already scheduled tasks) then
6:         if  $(\kappa_j > 0)$  then Generate random number  $q$ 
7:         if  $(q < P^1)$  then
8:           Add task  $j$  to line-of-work
9:           Update competence score of worker
10:        else
11:          Generate random number  $q$ 
12:          if  $(q < P^2)$  then
13:            Add task  $j$  to line-of-work
14:            Update competence score of worker
15:        Compute reduced cost of line-of-work
16:        if (reduced cost < 0) then
17:          Add line-of-work to the set of columns  $L$ 
18:        counter = counter + 1
19:      while (counter <  $U$ )
20: if (No line-of-work is added to the set of columns  $L$ ) then
21:   Repeat line 1-19 with  $P^2 = 0$  and  $U = 1$ 
22: #cycles = 0 Genetic algorithm
23: do
24:   Select two parents with same days-off pattern
25:   Apply uniform crossover
26:   for each task  $j \in J$  with  $d_j$  being a working day in the selected days-off pattern do
27:     Apply position-flip mutation with  $P^{mut}$ 
28:   Remove overlapping tasks
29:   Add line-of-work to population
30:   Compute reduced cost of line-of-work
31:   if (reduced cost < 0) then
32:     Add line-of-work to the set of columns  $L$ 
33:   #cycles = #cycles + 1
34: while (#cycles <  $F^{iter}$ ) and (No line-of-work added to set of columns  $L$ )

```

- A *genetic algorithm* that manipulates the lines-of-work derived via the single-pass heuristic. To this end, we use a binary encoding with a length equal to the number of tasks. Each position indicates whether the associated task is included in the line-of-work ('1') or not ('0'). Each evolutionary cycle, two lines-of-work, called parents, with the same days-off pattern are selected and a uniform crossover is applied, such that positions are copied from the one or other parent with equal probability. The newly created lines-of-work, called offspring, are subject to a mutation, which flips task positions from 0 to 1 or vice versa with a probability P^{mut} while the overlapping task modes are removed. The derived feasible columns with a negative reduced cost are added to the set of columns L at the end of each cycle. The process of creating offspring is repeated F^{iter} iterations or until a line-of-work with negative reduced cost is found.

5.4.3. Branching scheme

An optimal LP solution can be fractional and branching is then necessary to find a high-quality integer solution. We propose a variant of follow-on branching to generate a binary branching tree, creating two branches that impose restrictions on the task assignment variables x_{jw} . The branching scheme selects each time multiple tasks and imposes that all selected tasks are included into the same line-of-work or prevent that they are assigned to the same worker. To determine which tasks and worker are involved, the task-worker assignment with the largest fractional value S_{jw} , i.e. closest to 1, is chosen for branching and added to the set J_w^{br} . The task-worker assignment value for each task-worker combination is calculated in Eq. (39) by summing all the assignment

values of the columns for the considered worker that contain the examined task.

$$S_{jw} = \sum_{l \in L_w} \delta_{lj} u_{wl} \quad \forall w \in W, \forall j \in J \quad (39)$$

After the branching task j' and worker w' has been selected, we identify columns that contain task j' and add the other tasks included in these columns to the set $B_{j'}$. In order to determine the relevant set of tasks $J_{w'}^{br}$ for branching, we iterate over all tasks $j \in B_{j'}$ and add all those tasks j having the same task-worker assignment value as the branching variable, i.e. $S_{jw'} = S_{j'w'}$, to the set $J_{w'}^{br}$. If no such tasks exists, we again iterate over all tasks $j \in B_{j'}$ and look for the task that has the highest task-worker assignment value $S_{jw'}$. This task is added to $J_{w'}^{br}$. Branching is performed as follows. In the left branch, worker w' is forced to execute task j and all other selected tasks $j \in J_{w'}^{br}$, whereas the right branch prohibits that a worker is assigned to all tasks $j \in J_{w'}^{br}$. These branching decisions are added to the pricing problem via Eqs. (40) and (41), respectively.

$$\sum_{j \in J_{w'}^{br}} x_{jw'} = |J_{w'}^{br}| \quad (40)$$

$$\sum_{j \in J_{w'}^{br}} x_{jw'} < |J_{w'}^{br}| \quad (41)$$

6. Experimental analysis

This section provides computational and managerial insights into the impact of incorporating learning and training on the staffing decision. In Section 6.1, we introduce the test design and characteristics of the test instances. The computational performance and solution quality of the proposed procedure is assessed by making a comparison with alternative solution procedures in Section 6.2. Section 6.3 examines the benefits of learning and training under different conditions and the mutual differences between the different models. All experiments are conducted in Xcode 11.1, linked to the commercial solver Gurobi version 9.0.2. The tests are executed using an Intel Core i5 processor (1.60 GHz) and 8 GB of memory.

6.1. Test design

Test instances

As there are many possible applications for the staffing problem under study, we consider a generic problem definition and generate a set of instances based on the data generator of Krishnamoorthy and Ernst (2001) with the so-called tightness and a number of workers as input values. The tightness is defined as the aggregated duration of all tasks divided by the duration of the considered number of days times the input number of workers. However, these input values are only relevant to generate instances for the problem under study because we are minimising the number of required workers given the set of tasks. The data generator apportions tasks evenly over the different input workers and days. The duration of individual tasks is determined by randomly distributing the envisaged duration for all tasks based on the tightness value over the number of tasks per worker and day. The start times of tasks are defined based on their duration and the idle time, which is scheduled randomly between tasks. The instances are characterised by a planning horizon of seven days and are divided in three groups depending on the instance size relative to the number of tasks. We assume that each task should be staffed by exactly one worker ($R_j = 1$). This setting basically encompasses the splitting of activities, which possibly require multiple workforces, into unit tasks. The latter have a requirement of only a single worker per task and has been defined for following reasons, i.e. (i) its relevance to many application areas such as production, healthcare, retail, etc. as indicated in the study of Krishnamoorthy and Ernst (2001), which defined generic problem variants for the personnel task scheduling problem,

Table 3

Instance size characteristics of the test dataset.

Test instance	# instances	Input tightness	Input # workers	Avg # tasks	Avg task duration
Group A	20	0.30	5	70	216
Group B	20	0.30	10	140	216
Group C	20	0.30	20	280	216

and (ii) the research on learning and forgetting models focuses on the learning and forgetting of individual workers (see the review paper of Jaber (2014)), whereas the required assignment of multiple workers to activities complicates the determination of the efficiency the task is operated. Accordingly, we believe that setting $R_j = 1$ enables to evaluate the impact of training and learning on the staffing outcome in an unambiguous manner. Note that additional experimentation has been performed to examine the impact of having larger staffing requirements and validate the wide applicability of the presented research, which is described in Online Appendix A. The details of the instances can be found in Table 3.

The anonymous workforce is homogeneous in the sense that all workers have the same initial competence score ($i_{wg} = 0$) for each task class $g \in G$ at the start of the planning period. The hiring cost c_w of a worker is equal to 100. Hired workers can perform at most $m = 5$ working days in a period of seven days. This setting assuming a homogeneous workforce has been stipulated in view of its relevance to many application areas and its frequent presence in research studies involving personnel staffing. We refer to the literature overviews of Kim and Nembhard (2010) and Defraeye and Van Nieuwenhuyse (2016) and specific studies involving postal services (Bard et al., 2003) and project management (Maenhout & Vanhoucke, 2016), amongst other. Moreover, this assumption ensures the impact of training and learning on the staffing budget can be discerned more clearly and in a univocal manner. Note that additional experimentation has been performed considering the hiring of a heterogeneous workforce, which is described in Online Appendix B.

Gradual adaptation of workers' skill level and task duration

To model the relation between skill level and task duration t_{je} , tasks can be operated following eleven modes for each task (E_j) ranging from the baseline duration (t_j) to the shortest possible duration. The reduction in duration depends on the class the task belongs to. In total, four equally distributed task classes (G) are included and their maximum percentage reduction r_g amounts to 15%, 30%, 45% and 60%, respectively (Jaber, 2016). Furthermore, the default learning rate α is equal to 10%, the default forgetting rate β is equal to 5%, the default internal training improvement rate γ is equal to 15% and the default improvement rate ϵ of shadow training is equal to 13.5%. As concerns the off-the-job training, we suppose an improvement of 100% (χ), 48.5% (ψ) and 31.5% (ω) for external training that takes one day, half a day and a quarter of a day, respectively. Note that these learning rates have been set taking the mutual relative learning impact between learning and training methods into account. The parameters of the transformation function are set to 12.5 and 18 for parameter a and b , respectively. To facilitate shadow training, the tutor should have a competence score that is 15% (p) higher than the competence score of the apprentice. The cost of each type of external training is determined in such a manner that the cost per minute is the same, i.e. $c^{f1} = 4$, $c^{f2} = 2$ and $c^{f3} = 1$, i.e. a one-day off-the-job training is twice as expensive as a half-a-day training and four times as expensive as a quarter-of-a-day training. The values proposed in this section are considered ad-hoc 'educated guesses' and hypothesised to compose a generic test design. In real life, they are very depending on the characteristics of the involved company, the products and associated tasks (Hammer, 1954; Jaber, 2006). Note that, however, we investigate the sensitivity of these

settings in Section 6.3 in more detail so that the most suitable training method can be identified under varying conditions.

Methodology settings

Methodology settings have been determined based on preliminary computational experimentation. The corresponding insights validating the design choices and associated parameter settings relative to the constructive heuristic and multi-stage pricing procedure are described in Online Appendix C. Accordingly, the constructive heuristic composes repetitively a line-of-work for an additional employee as long as the number of uncovered tasks is higher than 50 (Q) (cf. Section 5.2). The number of lines-of-work that is fixed in further optimisation steps based on the solution generated via the constructive heuristic, is equal to zero, one and three (O) for groups A, B and C, respectively (cf. Section 5.3). Regarding the multi-stage pricing procedure (cf. Section 5.4), the threshold value LP^{imp} is set to 0.01 and at most 100 consecutive iterations of the heuristic pricing procedure (H^{max}) are allowed. Furthermore, the probabilities P^1 and P^2 relative to the heuristic pricing procedure are equal to 100% and 20%, respectively. The number of lines-of-work U constructed for a particular days-off pattern amounts to two. The genetic algorithm computes at most 20 iterations (F^{iter}) and considers a mutation probability (P^{mut}) equal to 10%.

Performance measures

To evaluate the proposed procedure, the following measures are reported:

Z^w	Number of required workers in the solution yielded via the proposed procedure
Z	Hiring cost of required workers in the solution yielded via the proposed procedure
t^{Tx}	Minutes of training relative to training type x in the solution yielded via the proposed procedure, e.g. for internal training $t^{To} = \sum_{w \in W} \sum_{j \in T^o} x_{jw} t_j$ (with x equal to o = internal training; a = shadow training; $f1/f2/f3$ = one-day/half-a-day/quarter-of-a-day training)
$\%Gap^1$	Gap between the calculated lower bound Z_{LB} (Section 5.1) and the yielded solution value Z ($\frac{Z - Z_{LB}}{Z}$)
$\%Gap^2$	Gap between the optimal solution value Z^* and the yielded solution value Z ($\frac{Z - Z^*}{Z}$)
$\%Gap^3$	Gap between the yielded solution value Z and the solution value Z_{UB} when no learning or training is applied ($\frac{Z_{UB} - Z}{Z_{UB}}$)
$\%Opt$	Percentage of instances for which a proven optimal solution has been yielded
$Time^T$	Total run time in seconds

6.2. Benchmark with other solution procedures

In this section, we benchmark the proposed method to alternative solution methodologies, known from the literature, for solving the staffing problem under study, i.e.

- **Mathematical programming (MIP):** This method solves the compact model Eq. (6)–Eq. (15) using a commercial MIP solver (e.g. Beaumont, 1997; Caprara et al., 2003). Note that MIP was not able to find a feasible solution for any instance of group B or C within the time limit. To accelerate its performance and provide a better benchmark, we narrowed its search space by restricting to some limited extent the number of workers that can be hired to reduce the number of symmetries without cutting off promising solutions.

- **Constructive heuristic:** This method conforms to the description in Section 5.2 encompassing the definition of smaller subproblems, which are solved in a sequential manner via mathematical programming by applying an employee-based decomposition (e.g. Chandrasekharan et al., 2020; Hojati, 2018).

The proposed method applies a dedicated branch-and-price procedure to find a high-quality solution for the problem under study, which is preceded by a constructive heuristic to improve the computational performance. To solve the problem using MIP or the proposed procedure, we have imposed a time limit of 3600, 14 400 and 36 000 s for groups A, B and C, respectively, based on preliminary experiments. Because mathematical programming fails to provide an accurate measure to assess the yielded quality due to its poor performance, comparison is made to an integer lower bound solution (cf. Section 5.1) to better assess the quality of the proposed procedure via metric $\%Gap^1$. Table 4 displays the benchmark results concerning only the staffing decision incorporating learning and forgetting. Computational performance for the staffing problem considering off-the-job training is similar to the presented results. Differences in solution quality between different forms of learning and training are discussed in Section 6.3.

Table 4 shows the computational performance of the proposed procedure in comparison to other procedures. In general, the proposed method realises the best solution quality as this method demonstrates the lowest gaps $\%Gap^1$ and $\%Gap^2$ for the different instance groups. MIP records the worst performance for all instance groups while requiring the largest run times. For the smallest instances of group A, MIP yields solutions having a significant gap of 19.52% in comparison with the lower bound. In addition, none of the 20 instances have been solved to optimality within the time limit and obtained solution values denote an optimality gap of 14.85%. For group B, MIP is able to solve 6 out of 20 instances to optimality after limiting the search space (cf. supra). The gap with the lower bound, however, is still substantial ($\%Gap^1 = 13.47\%$). No feasible solutions have been found within the time limit for the largest instances of group C. The constructive heuristic is able to reduce the optimality gap to 1.67% for instances of group A requiring an average run time of only 32.6 seconds. Also for the other instance groups, the constructive heuristic denotes smaller values for $\%Gap^1$ in comparison with MIP whereas the heuristic is several times faster (factor 120 and 86). The proposed procedure has clearly the best performance for all instance groups. For group A, the procedure is able to solve all instances to optimality, i.e. $\%Gap^2 = 0.00\%$, while the run time is significantly lower than MIP. For the other groups, the proposed procedure is able to find an optimal solution for 11 (5) out of 20 instances for group B (group C). Moreover, the lower bound gap $\%Gap^1$ attains a value of 5.56% and 5.75%, which is considerably low and leads to the presumption that (near-)optimal solutions have also been found for the other instances, for which optimality has not been proven, because these observed gaps are similar to group A. The run times of the proposed procedure are higher than the ones of the heuristic procedure but significantly better than run times yielded via mathematical programming using the MIP solver. Note that, dependent on the trade-off between solution quality and computational efficiency, the user may fix a larger number of lines-of-work in Step 3 of the proposed solution procedure, reducing the required run time, but probably – at the same time – worsening the obtained solution quality.

In Table 5, we evaluate the solution quality yielded by the different procedures by making a comparison with (i) an integer upper bound (Z_{UB}^w), i.e. the required number of workers without incorporation of learning and forgetting and initial expertise $i_{wg} = 0$ ($\forall w \in W, \forall g \in G$), and (ii) the integer lower bound (Z_{LB}^w) to have an idea on the minimum required number of workers when all operate continuously according to the highest competence score for all task classes. Comparison between Z_{UB}^w and Z_{LB}^w reveals that, on average, a reduction of 20% in staff is possible for the different instance groups when learning is considered in the staffing problem. Due to the inferior computational performance

Table 4

Computational benchmark to alternative solution methodologies.

	MIP			Constr Heur			Proposed		
	Group A	Group B	Group C	Group A	Group B	Group C	Group A	Group B	Group C
%Gap ¹	19.52%	13.47%	na ^a	6.67%	10.29%	12.60%	5.00%	5.56%	5.75%
%Gap ²	14.85%	7.59% ^b	na ^a	1.67%	8.07% ^b	11.15%	0.00%	0.93% ^b	0.00% ^b
%Opt	0%	30%	0%	90%	15%	0%	100%	55%	25%
Time ^T	3601.7	12 102.5	36 004.5	32.6	101.1	417.5	96.4	724.3	6776.7

^a Not applicable: No feasible solutions have been found within the time limit.^b %Gap² has been calculated only for solutions for which optimality has been proven within the time limit. Due to the line-of-work fixing ($|O| \geq 1$) (cf. Section 5.3), this can be proven only via MIP or via the closing of the optimality gap between the yielded upper bound and the lower bound (cf. Section 5.2).**Table 5**

Solution quality of the different solution methodologies.

	UB	Lower bound	MIP	%Gap ³	Constr Heur	%Gap ³	Proposed	
	Z_{UB}^w	Z_{LB}^w					Z^w	%Gap ³
Group A	6.05	4.85	6.05	0.00%	5.20	14.05%	5.10	15.70%
Group B	10.30	8.20	9.55	7.28%	9.15	11.17%	8.70	15.53%
Group C	18.00	13.70	na ^a	na ^a	15.70	12.78%	14.55	19.17%

^a Not applicable: No feasible solutions have been found within the time limit.

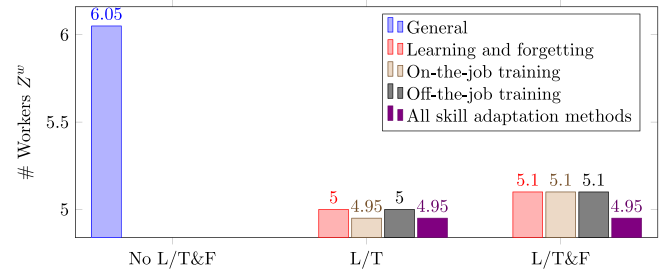
of MIP, the number of workers calculated via MIP is always worse compared to the proposed procedure and the constructive heuristic. The difference between the constructive heuristic and proposed procedure in calculated staff size for instances of group A is negligible as 5.20 versus 5.10 resources are staffed. However, the difference between these two methods in terms of yielded solution quality increases for larger instances; i.e. incorporating learning and forgetting improves the staff size with 15.53% and 19.17% for the proposed procedure and only 11.17% and 12.78% for the constructive heuristic. Absolute results reveal that the staff size yielded via the proposed procedure is considerably lower compared to the case without learning and forgetting and is fairly close to the maximum potential reduction in staff size that is realised when workers operate according to the largest skill level. These results lead to the conclusion that the proposed procedure is able to produce (near-)optimal solutions.

6.3. Managerial insights

This section provides insight into the applicability of different learning and training methods taking different parameter settings and conditions into account. In Section 6.3.1, we focus on the relevant parameters characterising the learning and training methods and the transformation function converting the worker's competence score into an efficiency value. Section 6.3.2 discusses the impact of the task characteristics on the performance of learning and training methods. In our test design, we analyse instance characteristics in a ceteris paribus manner, i.e. we vary one or multiple characteristics at a time leaving the other parameters unchanged to their default value indicated in Section 6.1. The experiments are only conducted for instances of group A, so that valid managerial insights can be yielded. For these instances, the proposed method finds optimal solutions when considering traditional learning and off-the-job training. When skill improvement can be realised via on-the-job training, mathematical programming is applied considering a time limit of 3600 s.

6.3.1. Impact of skill improvement and the transformation function

In the following, we analyse the benefits of gradually adapting the workers' skill levels on the staffing budget. First, we analyse the different learning and training methods individually, the impact of the forgetting effect, and the complementarity of different methods. Further, we study the effect of the skill improvement rates and the efficiency transformation function. The impact of different forgetting rates is discussed in Online Appendix D.

**Fig. 4.** The impact of the learning, training and forgetting on the staffing budget.

Benchmark comparison between learning and training methods

The different learning and training methods are compared in a general manner, whether or not combined with forgetting. Fig. 4 displays the required number of workers when (i) competence levels of workers cannot be adapted ('No L/T&F'), (ii) learning and training methods are applicable ('L/T'), and (iii) learning, training and forgetting are considered ('L/T&F').

The results reveal that incorporation of learning and training methods significantly reduce the required number of workers by more than one worker on average for these smaller-sized instances. The inclusion of on-the-job training leads to the lowest number of required workers because the planning of these training sessions is very flexible, so that these can be easily incorporated in the schedule of a worker between the other tasks that need to be executed. Moreover, as workers can more easily increase their efficiency level, they are capable of giving shadow training to other workers. Introducing the forgetting effect slightly increases the required number of workers for all individual learning and training methods as the forgetting effect gradually deteriorates the efficiency level of workers. This increase ranges between 0.1 and 0.15. The largest increase is denoted when additional skills are acquired via on-the-job training as on-the-job training sessions are shorter such that they only induce a slight increase in the skill level of the worker, which is more easily nullified by the forgetting effect. Regarding the complementarity of skill adaptation methods, results reveal that without forgetting the staffing budget does not improve when combining all skill adaptations methods together. However, methods are complementary when the forgetting effect is relevant. In contrast to the budget attained via individual methods, there is no increase in the number of required workers due to forgetting. Combining learning

and training methods provide so many alternatives to increase the workers' efficiency level such that a decrease in skill level can be easily compensated. This synergetic effect of learning and training has also been observed in the studies of Nembhard and Uzumeri (2000b) and Jaber et al. (2003), which examined to which extent training and deployment on top of learning contribute to reduce forgetting effects.

Impact of skill improvement rates Because the learning rate varies depending on the nature of tasks and industrial application (Dar-El et al., 1995), amongst other factors, different skill improvement rates are evaluated for each individual learning or training method. We consider in this assessment the forgetting effect for all methods. The impact on the required number of workers and the aggregated amount of training (in minutes) as indication for the training cost is displayed in Fig. 5 and Fig. 6, respectively.

Fig. 5 reveals improvements in the required number of workers for every learning and training method when the skill improvement rate increases. When these rates are increased to values larger than 30%, no further improvements are discerned. In general, traditional learning outperforms the training methods for all skill improvement rates. Inclusion of on-the-job training yields approximately the same number of workers, especially for moderate to large skill improvement rates. The good performance for on-the-job training can be explained by the fact that this training is characterised by a short duration and is planned in a flexible manner. Only when skill improvement rates are relatively low, relying on learning to adapt workers' skill levels yields a higher efficiency. The latter is explained by the fact that this form of learning happens automatically such that the worker can still benefit regularly from an increase in efficiency even when the improvement is only limited, while for the other forms of training explicit training tasks need to be scheduled, which consumes time and hampers productivity. When in the latter case the increase in efficiency is restricted, this increase does not fully compensate for the time spent on training activities away from the workplace, leading to a larger number of required workers. Fig. 6 shows that a considerable amount of internal training is inserted in the workers' lines-of-work. When the skill improvement rate is larger, this amount of internal training decreases, whereas the amount of shadow training slightly increases because workers possess more likely the necessary skills to educate other workers. However, as productive tasks are quite long, shadow training is not frequently done.

When skills can only be improved via off-the-job training, on the other hand, a larger number of workers is required taking the same skill improvement rate into account, which can be explained by the longer duration of external training during which workers cannot complete other productive tasks. In case only a small skill improvement can be realised via off-the-job training, the potential gains in efficiency do not outweigh the longer training duration, so that yielded solutions do not incorporate off-the-job training in the lines-of-work of the workers. However, when the skill improvement rate relative to off-the-job training increases and this type of training becomes attractive, the scheduled amount of training increases (cf. Fig. 6) and the resulting number of workers decreases. Once the skill improvement rate is higher than 30%, the number of workers cannot decrease anymore and less training is required to attain a sufficiently high skill level and the same efficiency.

The impact of the transformation function

To evaluate the impact of the relationship between learning/training and duration of tasks, we change parameters a and b , input to the transformation function $\phi(o_{ju})$ (Eq. (1)), and assess their impact on the required staffing budget. Because changing input parameter a has almost no impact on the required staffing budget for the different learning and training methods, the description of the experiment and associated results for parameter a can be found in Online Appendix D. Fig. 7 shows the duration averaged over all tasks for different competence scores by changing parameter b , leading to different shapes

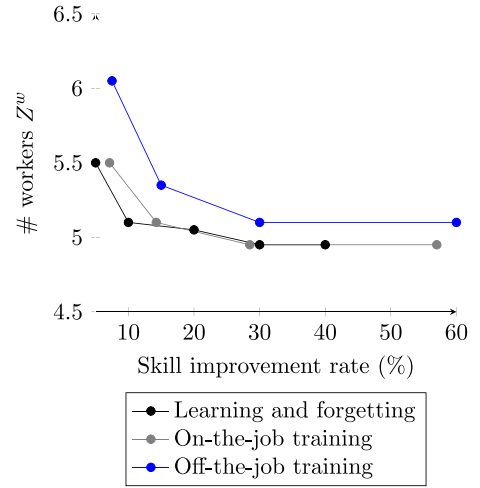


Fig. 5. The impact of the skill improvement rate on the required number of workers.

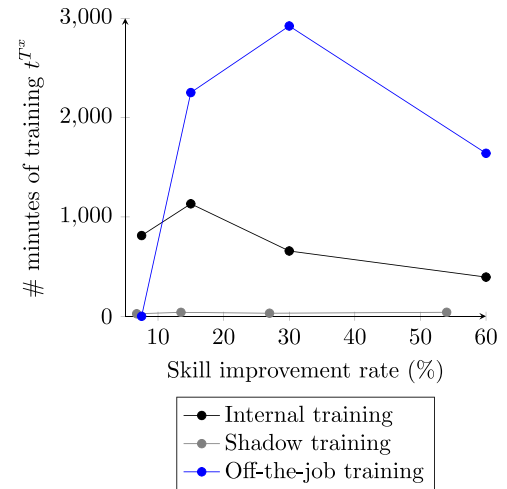


Fig. 6. The impact of the skill improvement rate on the amount of training.

of the transformation function and associated efficiency values. When changing parameter b , we observe divergent curves with approximate convex, concave and linear behaviour. Convex and linear curves are foremost applicable to cognitive tasks related to e.g. the software industry, which are characterised by a faster learning rate, whereas concave curves applies to motor tasks related to e.g. the production industry (Jaber et al., 2003). Note we assumed a maximum value of 0.5 as competence score (cf. Section 3), beyond which no further reduction in task duration is possible. Accordingly, this cut-off value is applied to the figure. If curves would be extended to a competence score of 1, they would all attain the same average minimal duration.

The effect of input parameter b can be clearly observed in Fig. 8. Especially when the value of b is low, the required number of workers is significantly higher than for other values, with up to 5.75 workers that are required. As can be seen in Fig. 7, a low value of b results in an approximate concave transformation function and only minor decreases in task duration can be realised, even if the workers' competence score level is high. Consequently, when $b = 4.5$, the average improvement in staffing budget thanks to skill adaptation is only 5.51% compared to the realised budget without learning or training (cf. supra). However, when the relationship is approximately linear ($b = 9$) or convex ($b = 18$), the staffing budget can be decreased with 14.33% and 15.70%, respectively, because of the larger possible reductions in task duration. Traditional learning is most susceptible to changes to parameter b .

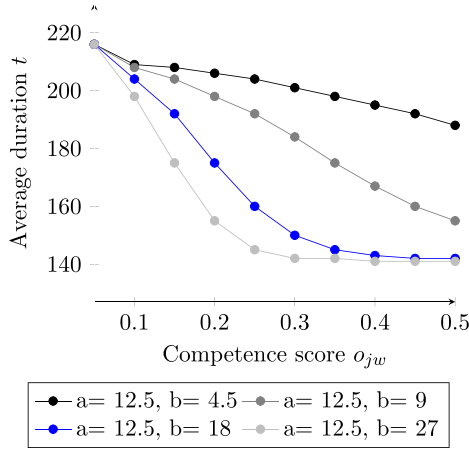


Fig. 7. The impact of input parameter b on the average task duration.

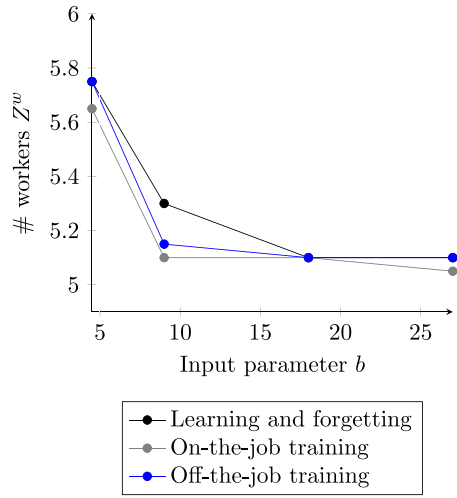


Fig. 8. The impact of input parameter b on the required number of workers.

Overall, we can conclude that the parameter b has a larger impact than parameter a , as the staffing budget is significantly impacted by the shape of the transformation function.

6.3.2. Impact of task set characteristics

In the following, we analyse the sensitivity of task set characteristics on the performance of the different learning and training methods. More precisely, we investigate the required skill heterogeneity indicated by the number of task classes and the number of tasks given a particular tightness value, which both have been demonstrated to impact the skill improvement and task efficiency (e.g. Jaber & Bonney, 1999). The impact of the maximum percentage reduction in task duration is discussed in Online Appendix D.

Required skill heterogeneity

This experiment varies the number of task classes between one and eight as a measure for the required skill heterogeneity. Fig. 9 reveals that results are different between learning and the training methods. Increasing the required skill heterogeneity from one to eight task classes leads to only a slight increase of maximal 0.05 and 0.1 resources for on-the-job and off-the-job training, respectively, which shows the suitability of training for a wide range of applications conforming to observations in literature (Ahadi & Jacobs, 2017; van der Klink & Streumer, 2002). Via learning, however, we observe a gradual increase with a larger number of task classes and a difference of 0.45 resources

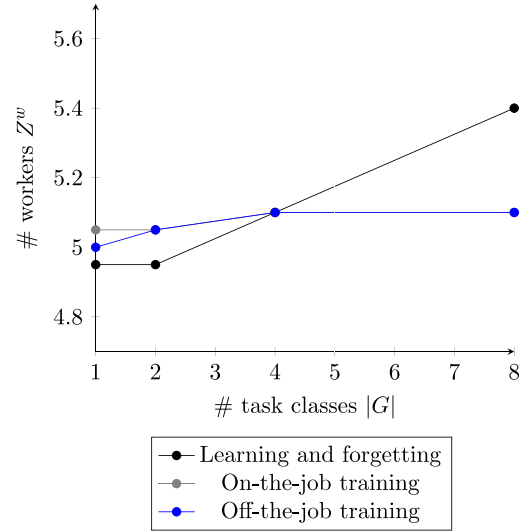


Fig. 9. The impact of the number of task classes on the staffing budget.

between the minimum and maximum skill heterogeneity. Moreover, when the number of task classes equals one or two, i.e. the required skill heterogeneity is low, lower staffing budgets are realised via learning. This is explained by the fact that each time workers execute a task, their skill level improves thanks to learning, so that the next task can be executed at a higher efficiency. In case of a larger required heterogeneity, a worker's efficiency rate for a particular task class increases more slowly as less tasks of the same task class are present and the time between these tasks is larger so that the forgetting effect is more prominent. These findings are consistent with the study of Jaber et al. (2003), which concluded that with an increasing task similarity the importance of training declines as learning or self-deployment provides satisfactory results.

The results related to training are explained as follows. A larger required skill heterogeneity expedites a larger degree of workforce training. However, it is not effective nor efficient that workers follow training related to all task classes, due to the large heterogeneity in tasks and limited number of tasks per class. Instead, every worker specialises rather in one or a few specific skill classes, enlarging the staffing budget only slightly. Workers are deployed to complete tasks for which their skill level has been augmented substantially via training. In this way, the impact of a larger required skill heterogeneity on the staffing budget is kept limited. When the required skill heterogeneity is lower, i.e. one or two task classes are discerned, the degree workers attend training activities is significantly lower. Only some workers will complete training to improve their skill level, which realises already a sufficient reduction in duration of tasks to attain the minimum staffing budget. A larger amount of training will increase unproductive time of workers, which is not compensated by an adequate further reduction in duration of tasks such that the staffing budget will increase.

Number of tasks

In this experiment, we vary the number of tasks per day to values of 5, 10 and 20 (or 35, 70, and 140 tasks per week), assuming the same tightness value of 0.3 and an input number of workers equal to 5 (cf. Section 6.1) as input to the problem generator of Krishnamoorthy and Ernst (2001). In this way, the average duration of considered tasks amounts to 360, 216 and 108 min. The results are displayed in Fig. 10. On-the-job and off-the-job training perform identical for different numbers of tasks considered. In the default case, considering 70 tasks per week (cf. Section 6.1), learning and training achieve the same performance.

Fig. 10 reveals that varying the number of tasks has a large impact for all skill improvement methods. As workers can execute only one

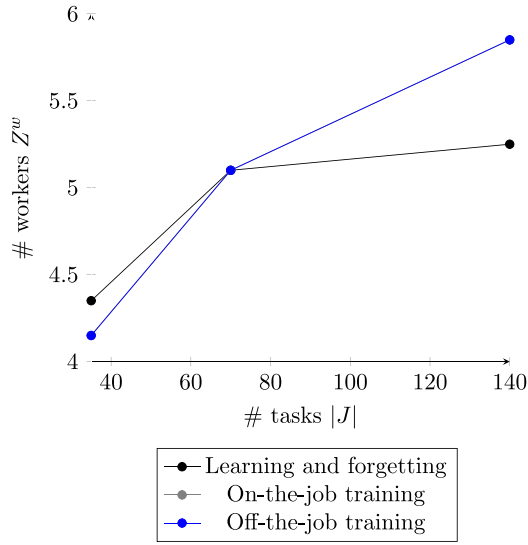


Fig. 10. The impact of the number of tasks on the staffing budget.

task at a time and the start times of tasks do not match perfectly, the number of required workers is larger when the number of tasks increases. These changes are more limited when traditional learning is considered caused by the learning potential. When fewer tasks are present, learning cannot be exploited that frequently to reduce the duration of tasks. When more tasks need to be executed, the learning potential is enlarged such that more tasks are carried out in modes with a shorter duration. The latter is conforming to the results of [Jaber and Bonney \(1999\)](#), which indicated that with learning it is recommended to perform smaller tasks more frequently. Consequently, the increase in the staffing budget as a result of learning is quite modest when doubling the number of tasks and equals only 0.15 resources. When considering on-the-job or off-the-job training, we observe the opposite phenomenon. When only few tasks need to be scheduled, fewer staff is required compared to traditional learning, which is thanks to following reasons, i.e. (i) it is easier to schedule training tasks between other tasks, and (ii) training leads to a larger increase in competence score such that the longer average duration of the tasks can be significantly reduced, compensating the time required for training. When there is a larger number of shorter tasks present, however, training realises a lower added value as it is more difficult to schedule training sessions between tasks and reductions in the duration of tasks are limited. Consequently, the average required staffing budget amounts to 5.85 workers for both on-the-job and off-the-job training, which is significantly higher than the budget realised when traditional learning is considered.

7. Conclusions

In this paper, we studied the personnel staffing problem that incorporates the gradual adaptation of workers' skill levels throughout the planning horizon. The contribution of this paper is threefold.

First, we proposed a new staffing model adapting the workers' skill levels in a dynamic manner, considering learning and forgetting of workers and a range of on-the-job and off-the-job training varieties. Due to the rapid pace of technological and global advances, employee training and development have become indispensable in business organisations, having a significant impact on their (financial) success through improving employee performance. Technological advancements and the competitive market have brought about the need for leveraging competences to perform specific tasks in a cost-effective manner. In contrast to static staffing models from literature that assume

a predefined skill mix and neglect the benefits of learning and training, we integrated these skill adaptation methods into the days-off scheduling and task assignment problem to determine the required staffing budget and training requirements more accurately. More specifically, we modelled the consequences for workers having a particular skill level, which is rarely done in the literature, via definition of multiple discrete modes for each task and ensured the selection of an appropriate mode based on the prevailing worker competence at the moment the task is executed, which determines the required duration to complete the task.

Second, we developed a novel procedure that hybridises principles of heuristic optimisation and mathematical programming, conforming to the literature. The proposed method starts with a constructive heuristic solving the longest-path problem per worker and fixes the best-fit lines-of-work for some of the workers. Subsequent optimisation is performed via a dedicated branch-and-price method that deploys follow-on task branching and multi-stage pricing to accelerate computation. Computational experiments have demonstrated the method outperforms a constructive heuristic and mathematical programming solving the compact formulation of the studied problem.

Third, we provide managerial insights into the desired deployment of learning and training methods and their impact on the staffing budget. Acknowledging the dynamic adaptation of workers' skill levels in the staffing decision provides opportunities to reduce operational costs on the long term, depending on environmental conditions relative to industry and/or company. All types of learning and training methods trigger improvements in the staffing budget. Apart from the learning rate, which may vary significantly depending on the nature of the task and industrial application, a determining factor is the shape of the transformation function between a worker's competency score and the task duration. Improvements in the staffing budget are considerable when the transformation function is approximately linear or convex as small improvements in a worker's competency lead to more significant reductions in the duration of tasks. Significantly lower improvements are discerned when the transformation function is concave. The former is mainly relevant for cognitive tasks, whereas the latter rather involves motor tasks. The complementarity of different learning and training methods shows, as combining them completely neutralises the negative effects of forgetting conforming to the literature. For each individual skill adaptation method, we can draw the following conclusions:

- Traditional learning, which is realised effortlessly via execution of tasks, leads to significant budget savings for any improvement rate and (maximum) reduction in task duration and is therefore widely applicable. However, in environments characterised by large forgetting rates, typically involving cognitive tasks, efficiency gains suffer and more formal training is recommended. Learning is more relevant when the number of tasks is larger and the required range of skills to execute tasks is limited. The smaller the number of tasks and the wider the range of skills that are required, the more formal training is relevant.
- On-the-job training requires the completion of (unproductive) training tasks, for which the allocation is organised in a flexible manner, compliant to the timing of other tasks. Accordingly, on-the-job training is able to realise considerable improvements in the staffing budget similar to traditional learning. The benefits for on-the-job training are robust to the required skill heterogeneity demonstrating its wide applicability.
- Off-the-job training requires a considerable investment of workers as they need to complete longer training tasks with little scheduling flexibility. Consequently, this type of training can only offer added value if workers' skill levels can be increased considerably and reductions in task duration are significant, i.e. especially for cognitive tasks. Moreover, in that case, off-the-job training is fairly robust to the effect of forgetting and the required skill heterogeneity. The latter is caused by the fact that personnel can specialise more easily in one or a few task classes, keeping the required number of resources low.

CRediT authorship contribution statement

Tessa Borgonjon: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Broos Maenhout:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cie.2023.109784>.

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